

Harsha Raj Kumar

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EDUCATION

Vanderbilt University M.S. Computer Science GPA: 3.6/4.0 Nashville, TN Seeking 2027 new-grad backend and platform engineering roles	Expected Apr 2027
Vellore Institute of Technology B.Tech. Computer Science GPA: 3.5/4.0 Chennai, India	May 2025

TECHNICAL SKILLS

Languages: C++, Java, Python, Go, TypeScript/JavaScript, SQL
Backend: Node.js/Express, FastAPI, Kafka/Redpanda, PostgreSQL/pgvector, Redis, REST APIs
Tools: Docker, AWS, GitHub Actions, Prometheus, React, PyTorch, SQLite

OPEN SOURCE

- PyTorch: contributed an accepted torch.unravel_index fix for a zero-dimension error case, with regression tests; landed in upstream core after maintainer review.

EXPERIENCE

Amplify GenAI Innovation Center, Vanderbilt University Software Engineer - Backend & Platform Systems - Developed REST and streaming APIs with FastAPI and Redis for Majik, supporting LLM tool calls and session state; added hybrid retrieval with PostgreSQL/pgvector. - Investigated save, delete, and sharing defects during Amplify UI testing, tracing failures to backend calls and exposed internal fields.	Aug 2026 - Present
VU-BEAM Lab, Vanderbilt University Systems & Software Engineer, Research Assistant Implemented ReCL contrastive pretraining and supervised fine-tuning in PyTorch, with ultrasound data loading and augmentation.	Oct 2025 - Aug 2026
Vanderbilt University Mathematics Department Software Engineer Developed the Proofdesk editor with React/Monaco and a Node/Express backend; continued the open-source project after the appointment.	Sep 2025 - May 2026

SELECTED PROJECTS

PulseStream TypeScript, Kafka/Redpanda, Redis, PostgreSQL Built an Express telemetry API and Kafka consumer with request idempotency and database conflict handling. A recorded local HTTP load test averaged about 3,991 responses/sec over 30 seconds at 50 connections; this measures ingestion acknowledgments.
Engram TypeScript, SQLite, ReAct Built a bounded tool-use loop and persistent memory engine with retrieval scoring, decay, entity associations, and structured traces. Compared retrieval strategies on a small synthetic fixture.
LSM-Tree C++ Project Lead and Maintainer Led review and integration of contributor-written storage internals. Authored architecture and benchmark documentation; reviewed WAL append-offset design and requested recovery regression tests.
Proofdesk React, Node/Express, Pyodide, Y.js Introduced browser-side PreTeXt/XML previews and collaborative editing integrations. Maintained the editor with contributors who expanded parsing and added security and queue-fallback fixes.

PUBLICATION & CERTIFICATIONS

- Co-author, "An Integrated GCN-GAT-AE Framework for Robust Anomaly Detection in Industrial IoT Environments," IEEE CICT 2025.

AWS Certified AI Practitioner, Jul 2026 | Microsoft Azure AI Fundamentals